

INTELLIGENCE GATHERING UPS AND INVERTERS

Customers' demand for increased reliability and zero downtime has led manufacturers to design more robust components and technologies for UPS and inverter systems. Some innovative technologies doing the rounds in the market lately can take the power electronics industry to the next level of cutting edge

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PAPERLESS WARRANTY

When a customer purchases a UPS or an inverter, he may face a lot of issues regarding warranty. Either the dealer or distributor does not hand over the warranty card to the customer or keep-



ing track of paper receipts becomes an encumbrance for the user. Sometimes, even the date on the warranty card is missing. To top it all, adopting unscrupulous means to keep customers in the dark concerning warranty details is also a common practice.

Now, UPS and inverter manu-

facturers offer a technology-enabled service intended to make warranty issues simpler for users, while also helping them to go green. It is called digital paperless warranty. Manufacturers are increasingly opting for this technology.

UPS and inverters are basically microcontroller-based systems. When you switch on the inverter or UPS for the first time, the microcontroller can do real-time data logging. The date of the first usage of the inverter or UPS gets logged into the memory of the microcontroller and after about six months or so, if there is a problem with the device, you can see the warranty information directly from the memory of the microcontroller, as it is equipped with a built-in recorder system. This takes care of all issues related to the warranty card and can check misuse of warranty as well.

INTELLIGENT CHARGING

By using a microcontroller, a battery charger can also be made intelligent. There is a charging section in the inverter and UPS which is connected to the battery. There is also an input and output section. Improvement is



taking place in all three of these sections. Today, manufacturers are substituting the bulky transformer-based charger with switch-mode power supply (SMPS) chargers. SMPS chargers are highly efficient, light-weight, low-cost with built-in fuzzy logic control for trickle charging. Fuzzy logic control aids in proper charging of the battery at a predetermined level. This concept is called intelligent battery charging.

Simple battery chargers do not possess the intelligence to handle different battery technologies or batteries with the same technology. They may leave the battery inadequately charged and can even pose serious safety hazards. In the intelligent battery charging technology, however, the microcontroller provides the intelligence to overcome these problems.

BATTERY EQUALISATION

This is also a built-in feature in UPS and inverters, which ensures that in

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